Certified Chart Residuals and Comparison Budgets for Complex Matrix Multiplication

A source-specific Ram Calculus study of M217

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Abstract

The assigned manuscript obtains its square matrix-multiplication claim through a rational, nearly harmonic six-probability chart, a polynomial barrier, and a lower-test inequality for an attainable tensor value. We separate these interfaces from the independent rectangular label-tree construction. Our local contribution is a quantitative comparison refinement: one-sided supporting-covector bounds become explicit polynomial panel inequalities, while an inset, a constant downward shift, and a quadratic curvature correction give a computable residual-tolerance frontier. With the source's other geometric and attainability premises retained, an exact worked choice tolerates a chart residual of 10^{-13} and still leaves a central utility excess greater than $1.1124 \cdot 10^{-6}$. This is a robustness statement, not a regenerated lower-degree chart or a better matrix-multiplication exponent. We prove the relevant finite enclosure identities, quantify chart-to-chart transfer, and preserve the exact entropy-equality condition needed for a dual-exponent witness. Two small original rectangular checkers are replayed separately from independent exact local checks; the full square computation is not replayed. Contribution assessment: C3/5, a notable source-specific residual-tolerance and signed comparison-certificate refinement, conditional on imported tensor and continuous-domain premises and with no new exponent claimed.

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1 Claims, conventions, and dependency boundaries

The target is the September 24, 2026 manuscript *Complex Matrix Multiplication Below* 2.258 *and Rectangular Bounds* [1]. Its source claims are

$$\omega_{\mathbb{C}} < 1129/500, \qquad \alpha_{\mathbb{C}} > .465, \qquad \omega_{\mathbb{C}}(1, .709, 1) < 2.092.$$

Here $w(k) = \omega_{\mathbb{C}}(1, k, 1)$ is an infimum of arithmetic exponents: an upper bound $w(k) \le t$ permits $O(n^{t+\epsilon})$ operations for every fixed $\epsilon > 0$. It is not an operation count with zero exponent slack, a bit-complexity bound, or a practical crossover estimate. Natural logarithms are used throughout this report.

The local theorems below are proved here. When instantiated with the source's chart, they import its stated finite outputs, geometric enclosures, and attainable-value lower-test theorem. Reading the complete scientific source is not a second proof of every tensor construction or every analytic estimate. We do not promote a coordinate change or a polynomial sign test into a new tensor algorithm.

1.1 What the source actually constructs

Its basic six-term tensor is

$$T = \sum_{i+j+k=2} x_i y_j z_k, \qquad 0 \le i, j, k \le 2.$$

The source supplies a border-rank-three identity. It separates two-sided-readable labels by a charged group tensor, leaving dot-product factors. Previously produced factors remain separate; later decisions read original direct labels, not these factors' internal indices. Exact conditional types make auxiliary supplies and output dimensions independent of the direct history.

A square construction has six physical orientation banks, each with the same canonical source law. With n original occurrences, L nonzero direct outputs, auxiliary rank budget R_{aux}, and common matrix-product volume v, its score is

$$f_\beta = \frac{\log L - \log R_{\text{aux}} + (\beta/3)\log v}{n}. \tag{1.1}$$

The source defines F_β by the Euclidean closure of finite law–score pairs. It proves concavity, continuity, and $0 \le F_\beta(\alpha) \le H(\alpha) \le \log 6$. The closure contains limits of finite constructions, not an infinite composition applied to a finite tensor.

Source module	Retained obligation
`algebra.tex`	Local group projections; no cross terms; canonical bank histograms; nonzero supported words; auxiliary charges.
`values.tex`, `ap.tex`	Closed attainable values; conditional recompilation; every-permitted-path control; finite tree compilation; lower-test inequality.
`certificate-w.tex`	Attainable W seed, actual finite menu laws, rational correction, and interpolation over the whole shell.
`certificate.tex`	Exact polynomial recurrence; full/reference differences; geometry, interior subdivision, face/cap boundary, positive inset.
`rectangular/`	Separate cost witnesses, readable trees, three edge rates, completion, exact flatness, and finite-family support optimization.
`conclusion.tex`	Strict utility selects a finite construction; asymptotic sum inequality uses its full auxiliary charge.

At $\beta = 1129/500$, the sufficient final finite inequality is

$$L^{1500} v^{1129} > (3^n R_{\mathrm{aux}})^{1500}.$$

Neither this report's tests nor the original numerical checkers exhibit all of the tensor maps whose existence is used at this final step. The source also distinguishes two fixed strict rank witnesses, which give finite-exception characteristic transfer, from accuracy-dependent dual witnesses, which do not give one common exceptional prime set. Those distinctions are retained.

1.2 Current public context and original private input

The SODA 2024 primary publisher statement of Vassilevska Williams, Xu, Xu, and Zhou gives $\omega \le 2.371552$ and $\alpha \ge .321334$ [2]. Dupont et al.'s August 2026 preprint reports $\omega < 2.371177$; its Theorem 1 makes feasibility the interface to the exponent, and its Section 4 describes a separate rational verification [3]. We inspected those statements and the verification description, not its full optimization proof or code. The source's baseline explicitly uses versioned tables, not every possible untabulated choice in those methods. Public repository formalization descriptions encountered during the search are not independent Lean executions here. No claim about community acceptance or global priority is inferred from a repository description.

The public overview also lists the separate October 2 companion *An Upper Bound of* $9/4$ *for the Matrix Multiplication Exponent*, claiming $\omega_{\mathbb{C}} \le 9/4$ [5]. Only that overview entry and the companion's public title/metadata were inspected here, not its proof. M217 remains the fixed September 24 source; our robustness theorem is not a new best-exponent claim.

Fresh local reading of the original Ram volume 16 clearance chapter [6] supplies the exact power-to-Bernstein proof, convex-hull enclosure, subdivision identity, and the separation of a polynomial enclosure from a nonpolynomial remainder. These are classical tools, not new Ram discoveries. Their use here is in the specific six-state chart and its supporting-plane error. No private text was sent to public search services.

2 The finite chart and its actual differential defect

Set $\zeta = e^{2\pi i/3}$, $R = 31/25$, and use physical coordinates $z = x + iy$. The real polynomial P defines

$$r_i = \frac{1 + \Re(\zeta^i z)}{3}, \quad p_i = P(\zeta^i z), \quad E_i = r_i + 2p_i - \sum_j p_j, \quad \alpha = (p_0, p_1, p_2, E_0, E_1, E_2). \tag{2.1}$$

All indices are modulo three. The coordinates sum to one; the grade mean is $\mu_i = 1 - r_i$. It is an affine invertible coordinate on the mean plane. Positivity, however, is a separate domain condition.

For $D_i = r_i(1-r_i)+2p_i$, the covariance matrix in (x,y) is

$$C = \begin{pmatrix} 9D_0 & 3\sqrt{3}(D_1-D_2) \\ 3\sqrt{3}(D_1-D_2) & -3D_0+6D_1+6D_2 \end{pmatrix}, \qquad \mathcal{L} = C : D^2. \tag{2.2}$$

This is the full contraction, not the stochastic generator divided by two. Indeed, for the centered grade ξ, $\sum_i \xi_i = 0$, so $\mathbb{E}\xi_i\xi_j = (D_k - D_i - D_j)/2$. The coordinate increment is $-3\xi_0 + i\sqrt{3}(\xi_1 - \xi_2)$, which gives (2.2). With

$$d = (D_0+D_1+D_2)/3, \qquad f = (D_0 + \zeta D_1 + \zeta^2 D_2)/3,$$

the same operator is

$$\mathcal{L}u = 36(du_{z\bar{z}} + fu_{zz} + \bar{f}u_{\bar{z}\bar{z}}), \qquad \mathcal{L}|z|^2 = 36d. \tag{2.3}$$

The last identity will pay for the local curvature correction.

2.1 Finite recurrence, not an infinite convergence assumption

For $Q = 2^b$, put

$$(Z_0, \dots, Z_7) = (18623, -2696, -2677, 194, 393, 17, 1, 8)$$

and $Z_k = 0$ for $k \geq 8$. The pure coefficients are $A_{0k} = A_{k0} = \lfloor QZ_k/10^5 + 1/2 \rfloor$. Let $c_{00} = 8, c_{10} = c_{01} = 2, c_{20} = c_{02} = -1, c_{11} = -2$, with the other c's zero, and $C_{ij} = 72A_{ij} + Qc_{ij}$. For $1 \leq i \leq j$, define

$$S_{ij} = \sum_{\substack{0\leq x<i,\ 0\leq y<j \\ (x,y)\neq(0,0)}} C_{xy} A_{a,c} w_h(a,c), \tag{2.4}$$

$$h \equiv 1+y-x \pmod 3, \quad 0 \leq h \leq 2, \quad a = i-x-1+h, \quad c = j-y+1-h,$$

$$(w_0, w_1, w_2) = (c(c-1), ac, a(a-1)), \quad A_{ij} = \left\lfloor -\frac{S_{ij}}{C_{00}ij} + \frac{1}{2} \right\rfloor, \quad A_{ji} = A_{ij}.$$

All right-hand coefficients have smaller total degree. The full and reference polynomials are, respectively,

$$P_{N,b} = Q^{-1} \sum_{i+j\leq N} A_{ij} z^i \bar{z}^j, \qquad P = P_{780,500}, \quad P^{\mathrm{ref}} = P_{420,224}.$$

The recurrence fixes both coefficients and rounding convention. It does not define a freely fitted polynomial.

Writing $\mathcal{C} = 36[r_0(1-r_0) + 2P_{N,b}]$, its residue classes of exponent difference give $\mathcal{L} = \mathcal{C}^{[0]}\partial_z\partial_{\bar{z}} + \mathcal{C}^{[2]}\partial_z^2 + \mathcal{C}^{[1]}\partial_{\bar{z}}^2$. For $i, j \geq 1$, $i+j \leq N$, the coefficient of $z^{i-1}\bar{z}^{j-1}$ in $\mathcal{L}P_{N,b}$ is exactly

$$Q^{-2}(C_{00}ijA_{ij} + S_{ij}). \tag{2.5}$$

Derivative factors annihilate the apparently out-of-range terms; the constant covariance coefficient supplies the displayed pivot. Nearest rounding gives absolute defect at most $C_{00}ij/(2Q^2)$. The source's positive coefficient sums also bound all products above the recurrence degree. A tiny low-degree defect alone would not do so.

For $F = \sum f_{ij} z^i \bar{z}^j$, the exact finite majorant $\|F\|_{\infty,|z|\leq R} \leq \sum |f_{ij}| R^{i+j}$ includes every coefficient. Differentiation costs degree $k = i+j$ and $k(k-1)$ for first and second unit real directional derivatives. Applying these identities to coefficient *differences before absolute values* is the correct way to compare P and P^{ref}. The source reports

$$\delta_0 < 3.517 \cdot 10^{-17}, \quad \delta_1 < 1.233 \cdot 10^{-14}, \quad \delta_2 < 4.327 \cdot 10^{-12}, \tag{2.6}$$

for the value, gradient, and Hessian differences on the entire disk. These three numbers are imported numerical outputs, not regenerated here.

3 Signed supporting-covector certificates

The source lower-test theorem says that, at an interior lower contact of ψ with $F_\beta \circ \alpha$,

$$\mathcal{L}\psi \le g \cdot \mathcal{L}\alpha - 1 \tag{3.1}$$

for every supporting supergradient g of F_β. Its proof uses finite AP count trees and the every-permitted-path property; it is not a consequence of a small polynomial residual. Only an *upper* estimate for $g \cdot \mathcal{L}\alpha$ is needed to contradict (3.1).

Proposition 3.1 (Sharp relaxed one-sided debt)**.** *Let $\alpha \in (0,1)^s$, $\sum_i \alpha_i = 1$, and $r \in \mathbb{R}^s$, $\sum_i r_i = 0$. Let h be concave on the full simplex, $0 \le h \le K$. Normalize a supporting supergradient by $\alpha \cdot g = h(\alpha)$. Then*

$$g \cdot r \le Km, \qquad m = \max_i r_i/\alpha_i \ge 0. \tag{3.2}$$

This is the exact supremum over the relaxed constraints $g_i \ge 0$, $\alpha \cdot g \le K$. If verified vertex bounds $h(e_i) \ge b_i \ge 0$ are supplied, then

$$g \cdot r \le b \cdot r + (K - b \cdot \alpha)m. \tag{3.3}$$

The latter is sharp for the corresponding relaxation $g_i \ge b_i$; it requires $K \ge b \cdot \alpha$.

Proof. Adding a constant vector to g permits the normalization and leaves $g \cdot r$ unchanged. The supporting plane is now $q \mapsto g \cdot q$. At e_i it gives $g_i \ge h(e_i) \ge 0$. Also $\sum_i \alpha_i(r_i/\alpha_i) = 0$, proving $m \ge 0$. Hence $g \cdot r \le m \sum_i \alpha_i g_i \le Km$. For an index attaining m, put $g_i = K/\alpha_i$ there and zero elsewhere to attain the relaxed optimum. With vertex information apply the same argument to $g - b$; put all of its mass on an index attaining m to prove sharpness. □

This is elementary linear optimization, credited as such. A maximizing relaxed vector need not be an actual supergradient of the tensor value. For M217 the entropy cap forces $h(e_i) = 0$; no positive vertex bonus is available without different, genuinely established supporting constraints. Mean-affine gauge freedom cannot be used to invent such information.

3.1 The three-residual structure

Let $h_i = \mathcal{L}p_i$ and $H = h_0 + h_1 + h_2$. Since each r_i in (2.1) is affine,

$$r := \mathcal{L}\alpha = (h_0, h_1, h_2, 2h_0 - H, 2h_1 - H, 2h_2 - H). \tag{3.4}$$

It has zero total and zero grade mean. It is generally not six independent errors. In particular $|h_i| \le \epsilon$ implies $|r_{p_i}| \le \epsilon$, $|r_{E_i}| \le 3\epsilon$. Taking $K = 2 > \log 6$ and $\alpha_i \ge \eta$ recovers the source's conservative $6\epsilon/\eta$. The factor three is attainable in this residual model, so it cannot simply be replaced by one.

Example 3.2 (A source-shaped asymmetric debt)**.** Take $p_i = 1/10$, $E_i = 7/30$, and $(h_0, h_1, h_2) = (-2, 1, 1)e$, $e > 0$. The residual is $(-2, 1, 1, -4, 2, 2)e$. The largest positive ratio is $10e$, whereas the largest absolute ratio is $20e$. For $K = 2$, the one-sided relaxed debt is $20e$, half the absolute-value estimate $40e$. This is an exact law/residual example compatible with (3.4), not a claim that the source polynomial has these derivatives at a point.

Corollary 3.3 (Denominator-free local test)**.** *On a positive chart with $0 \le F_\beta \le K$, it suffices for strict comparison of a C^2 function ψ that, for every coordinate i and every interior point,*

$$G_i := \alpha_i(1 + \mathcal{L}\psi) - K(\mathcal{L}\alpha_i) > 0, \tag{3.5}$$

and that $\psi \le F_\beta \circ \alpha$ on the boundary. For polynomial α, C, ψ, these are polynomial tests.

Proof. Divide by the positive α_i, take the maximum, and apply Proposition 3.1. If $\psi - F_\beta \circ \alpha$ had positive maximum c, it would occur inside; $\psi - c$ would be a lower test contradicting (3.1). □

This avoids separately maximizing six ratios, the curvature, and their absolute errors. It also permits a different active coordinate in different panels. It does not dispense with positivity of the law: a panel meeting an $E_i = 0$ face needs an inset, an interior exhaustion argument with proved boundary limits, or a separate boundary analysis. No division by a zero face coordinate is permitted.

4 Trading boundary and central slack for residual tolerance

The source's small residual is much smaller than required by its comparison argument. Merely observing slack would be a minor remark. Here we determine how much of it can be exchanged, with every loss charged in the source's original coordinates.

4.1 Uniform inset costs from the source data

The following are imported continuous-domain bounds from `certificate.tex`, not new outputs of a replayed square checker:

$$\|\nabla p_i\| < .592, \quad \|\nabla E_i\| < G = 2.11, \quad -\partial_r E_1 > s = .144 \quad (E_1 \le .01).$$

In the fundamental sector, the inactive coordinates obey $p_i > .0502926$, $E_0, E_2 > .0504548$; the other sectors follow by symmetry. The source's original boundary has $F_\beta \circ \alpha \ge \varphi + M$. Its interior and centre satisfy

$$\begin{aligned} M = 5.5 \cdot 10^{-6}, \qquad A = 7.094 \cdot 10^{-6}, \\ 1 + \mathcal{L}\varphi > \delta = 9.8 \cdot 10^{-8}, \qquad \varphi(0) - \log 3 > A. \end{aligned} \tag{4.1}$$

All finite decimals in this report denote exact rationals.

Lemma 4.1 (A uniform, larger inset window)**.** *Retain the source's domain, root uniqueness, face-sample band, derivative bounds, and finite arithmetic enclosures. For every $0 < \eta \le \eta_0 = 10^{-7}$, move each active endpoint inward to $E_i = \eta$, leaving an earlier cap endpoint unchanged. The resulting compact inset contains the origin, all its probabilities are at least η, and*

$$F_\beta(\alpha(z)) - \varphi(z) \ge M - L_b \eta \quad (z \in \partial\Omega_\eta), \qquad L_b = \frac{KG/m + V}{s} = \frac{10925}{18}, \tag{4.2}$$

where $K = 2$, $m = .05$, and $V = 3$. Moreover

$$\mathcal{L}|z|^2 > \kappa = 17.44. \tag{4.3}$$

Proof. The removed radial length is at most η/s. The inactive lower bound remains above $.0502926 - G\eta_0/s > .05$; the active coordinate increases. The source's radial monotonicity and positive values at the origin therefore give precisely the same inset construction on this larger window. This is an argument uniform in η, not an optimization over an unquantified limiting assertion.

For original and inset endpoint laws a and a', respectively, set

$$\lambda = \max_{a_i > 0} (1 - a_i'/a_i)_+.$$

The active coordinate contributes zero, and $\lambda \le G\eta/(sm) < 1$. If $\lambda > 0$, the vector $q = (a' - (1 - \lambda)a)/\lambda$ is a probability law. Concavity, $F_\beta(q) \ge 0$, and $F_\beta(a) \le K$ imply $F_\beta(a') \ge F_\beta(a) - K\lambda$.

If $\lambda = 0$, equality of total masses gives $a' = a$. This also covers an old cap endpoint whose active probability was already positive but below η.

We must justify $V = 3$ on the enlarged radial segment. The source's sample spacing is $h = \pi/(3 \cdot 65536)$. Its root Lipschitz constant 18.2, root error $2 \cdot 10^{-9}$, and enlarged displacement give polar-coordinate distance

$$d_* \le 19.2h + 2 \cdot 10^{-9} + \eta_0/s < .00031.$$

At the applicable face samples, the first three derivative bounds, including arithmetic error, are $2.902, 79.002, 1999.002$; fourth derivatives are bounded by $435 \cdot 10^6$. The source's band includes cap endpoints where the inset is active. Taylor's theorem along the polar segment yields

$$|\partial_r \varphi| < 2.902 + 79.002 d_* + \frac{1999.002}{2} d_*^2 + \frac{435 \cdot 10^6}{6} d_*^3 < 3.$$

Thus the barrier loss is at most $V\eta/s$. Adding the two losses proves (4.2); unchanged cap points have no transfer loss.

Finally the source's truncated centre values give $d \ge .488$, and its full-to-centre changes in every D_i are below .00334. Consequently $d > .48466$. Equation (2.3) gives $\mathcal{L}|z|^2 > 36(.48466) = 17.44776 > 17.44$. This uses a trace bound, not a replacement for the independently needed positive covariance and all-domain geometry. □

Theorem 4.2 (A three-budget residual certificate)**.** *Assume Lemma 4.1, the continuous attainable-value lower-test premise (3.1), and the rotational chart identities. Suppose $|\mathcal{L}p_i| \le \epsilon$ on the original domain. Choose $0 < \eta \le \eta_0$, $c, t \ge 0$ such that*

$$L_b\eta - M \le c, \qquad \frac{6\epsilon}{\eta} < \delta + \kappa t, \qquad c + R^2 t < A. \tag{4.4}$$

Then the corrected barrier

$$\psi(z) = \varphi(z) - c - t(R^2 - |z|^2) \tag{4.5}$$

satisfies $F_\beta \circ \alpha \ge \psi$ on $\overline{\Omega}_\eta$, and

$$F_\beta(\alpha(0)) - \log 3 > A - c - R^2 t > 0.$$

Proof. On the boundary, $R^2 - |z|^2 \ge 0$, so (4.2) and the first budget give $\psi \le F_\beta$. The constant c has no curvature cost; the quadratic correction increases $\mathcal{L}\psi$: $1 + \mathcal{L}\psi > \delta + \kappa t$. The chart identities give $|\mathcal{L}E_i| \le 3\epsilon$. Since all probabilities exceed or equal η, Proposition 3.1, or its absolute-value consequence, gives $g \cdot \mathcal{L}\alpha \le 6\epsilon/\eta$. If $\psi - F_\beta \circ \alpha$ had positive maximum inside, subtracting that maximum from ψ would produce a lower test with

$$1 \le g \cdot \mathcal{L}\alpha - \mathcal{L}\psi < 6\epsilon/\eta + 1 - \delta - \kappa t < 1,$$

a contradiction. Evaluation at the origin and the final budget finish the proof. Compactness supplies the maximum; no differentiability of F_β is assumed. □

Example 4.3 (An exact, source-calibrated tolerance)**.** Take

$$\epsilon = 10^{-13}, \qquad \eta = 10^{-8}, \qquad c = 6 \cdot 10^{-7}, \qquad t = 3.5 \cdot 10^{-6}.$$

The three retained slacks are, exactly,

$$\begin{aligned} M - L_b\eta + c &= \frac{11}{360000000} > 0, \\ \delta + \kappa t - 6\epsilon/\eta &= 1.138 \cdot 10^{-6} > 0, \\ A - c - R^2 t &= 1.1124 \cdot 10^{-6} > 0. \end{aligned}$$

The first is about $3.0556 \cdot 10^{-8}$. This allowed residual is over $3 \cdot 10^8$ times the source's reported ceiling $3.329 \cdot 10^{-22}$. The comparison is between sufficient tolerances, not between measured errors, polynomial degrees, or running times. The source's actual chart already passes with a much smaller residual and a larger uncorrected central margin.

4.2 The exact frontier of this scalar correction family

Proposition 4.4 (Optimal budget allocation)**.** *For fixed $\epsilon \geq 0$, the infimum central charge at inset η in this three-budget method is*

$$J_\epsilon(\eta) = (L_b\eta - M)_+ + \frac{R^2}{\kappa}(6\epsilon/\eta - \delta)_+. \tag{4.6}$$

If $J_\epsilon(\eta) < A$, the strict conditions in (4.4) can be satisfied with arbitrarily small additional charge. The supremum tolerable residual at that inset is

$$\mathcal{E}(\eta) = \frac{\eta}{6}\left[\delta + \frac{\kappa}{R^2}(A - (L_b\eta - M)_+)\right], \tag{4.7}$$

on $0 < \eta \leq \eta_0$ with $(L_b\eta - M)_+ < A$. Writing

$$B = A + M + \frac{\delta R^2}{\kappa}, \qquad \eta_B = M/L_b, \qquad \eta_v = B/(2L_b),$$

the source constants satisfy $\eta_B < \eta_v < \eta_0$ and $L_b\eta_v - M < A$. Thus their maximum is

$$\mathcal{E}_{\sup} = \frac{\kappa B^2}{24 L_b R^2} \quad \text{at } \eta = \eta_v. \tag{4.8}$$

The strict endpoint is not asserted to be admissible.

Proof. The smallest nonnegative constant shift is $c_0 = (L_b\eta - M)_+$. The infimum nonnegative curvature is $t_0 = (6\epsilon/\eta - \delta)_+/\kappa$. If necessary, increase t_0 slightly to make the middle inequality strict; $J_\epsilon < A$ leaves room. This proves (4.6). Conversely every permissible choice pays at least J_ϵ. Eliminating t under $c + R^2 t < A$ gives (4.7). Below η_B it is increasing and linear in η. Above η_B it is the concave quadratic $\kappa\eta(B - L_b\eta)/(6R^2)$, whose vertex gives (4.8). The stated rational inequalities check that this vertex lies in the permitted part of the window. □

Numerically $\mathcal{E}_{\sup}$ lies between $1.23 \cdot 10^{-13}$ and $1.24 \cdot 10^{-13}$. For a prescribed retained central excess $\gamma \in (0, A)$, replace A by $A - \gamma$ throughout the frontier and recheck the location of the vertex. This elementary constrained optimization is classical. Its contribution here is the explicit coupling to the source's actual boundary-dilution loss, enlarged inset window, covariance trace and utility slack. It is optimal only among the declared constant-plus-radial-quadratic corrections and these scalar bounds. Panelwise signed tests may do better; no optimality of the actual barrier or attainable value is claimed.

5 Whole-panel certification and chart transfer

The signed tests (3.5) preserve correlations between probability, curvature and residual. A centre evaluation with a floating tolerance is not a certificate for an entire panel. Here is an exact alternative interface, derived from the original clearance proof [6].

Lemma 5.1 (Finite Bernstein certificate)**.** *On a rectangle affinely identified with $[0,1]^2$, let $q(s,t) = \sum_{i=0}^m \sum_{j=0}^n a_{ij} s^i t^j$. With $B_k^m(s) = \binom{m}{k} s^k (1-s)^{m-k}$, its coefficients in the tensor Bernstein basis are*

$$b_{kl} = \sum_{i \leq k,\ j \leq l} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{l}{j}}{\binom{n}{j}}. \tag{5.1}$$

Consequently $\min b_{kl} \leq q \leq \max b_{kl}$. If a true panel expression is $q + e$ with $|e| \leq \rho$, then $\min b_{kl} > \rho$ proves its strict positivity everywhere. Rational subdivision produces exact child coefficients and does not alter the polynomial.

Proof. For $0 \le i \le m$,

$$\sum_{k=i}^{m} \frac{\binom{k}{i}}{\binom{m}{i}} B_k^m(s) = s^i \sum_{k=i}^{m} \binom{m-i}{k-i} s^{k-i}(1-s)^{m-k} = s^i.$$

Multiply the two one-dimensional identities and collect coefficients. The basis is nonnegative and sums to one, proving the hull bound. For a split at $u \in [0,1]$, repeatedly replace adjacent coefficients b_k, b_{k+1} by $(1-u)b_k + ub_{k+1}$. The binomial identity shows that the two boundary diagonals of this triangular array are the coefficients of the restrictions to $[0,u]$ and $[u,1]$. Applying this separately in both variables proves subdivision. The additional remainder is charged after enclosure, as claimed. □

For example, on $[-1,1]^2$, $q = 1 + x^2 + y^2$ has a useless constant-minus-absolute-tail bound -1. On the four rectangles cut by $x = 0$, $y = 0$, its degree-$(2,2)$ Bernstein coefficients have minimum 1. This illustrates a classical enclosure advantage, not evidence that the full M217 panels become cheaper. An unresolved leaf must be subdivided or rejected. Positivity at sampled points cannot replace the hull test. For a polynomial with a strictly positive minimum on a compact rectangle, sufficiently fine subdivision succeeds: Taylor expansion on a panel of diameter h and (5.1) bound every coefficient's difference from its corner value by $O(h)$, with a constant fixed by the polynomial. A polynomial vanishing on a face needs factoring or a separate non-strict argument; this termination statement gives no uniform complexity as the positive minimum tends to zero.

5.1 Exact fields, analytic remainders, and correlated error

In physical coordinates the rotations involve $\sqrt{3}$. Thus a rational certificate does not mean that every Cartesian coefficient is rational in x, y. One may work in the ordered quadratic field $\mathbb{Q}(\sqrt{3})$, with certified rational brackets for comparisons. Alternatively put $z = x + i\sqrt{3}\,u$. The chart polynomials then have rational coefficients, and the covariance becomes

$$C_{x,u} = \begin{pmatrix} 1 & 0 \\ 0 & 1/\sqrt{3} \end{pmatrix} C_{x,y} \begin{pmatrix} 1 & 0 \\ 0 & 1/\sqrt{3} \end{pmatrix}.$$

The disk is $x^2 + 3u^2 \le R^2$. This is a coordinate change of the operator and domain, not an isometry of derivative norms. Only rectangles contained in, or rigorously clipped to, the actual positive domain may be certified. Polar panels instead require the source's trigonometric Taylor remainders; a finite Taylor array is not the original function without its remainder.

Proposition 5.2 (A product-aware transfer budget)**.** *Suppose a panel has approximate quantities $a, b, \widetilde{r}$ for $\alpha_i, 1 + \mathcal{L}\psi, \mathcal{L}\alpha_i$, with pointwise errors $e_\alpha, e_B, e_r \ge 0$. Then*

$$|G_i - (ab - K\widetilde{r})| \le |a|e_B + |b|e_\alpha + e_\alpha e_B + Ke_r. \tag{5.2}$$

Uniform upper bounds for the right side, combined with Lemma 5.1, yield a true full-panel test.

Proof. Write the two exact factors as $a+\Delta a, b+\Delta b$. Their product difference is $a\Delta b + b\Delta a + \Delta a\Delta b$. The triangle inequality gives (5.2), including the often omitted product of errors. Add K times the residual error. □

This preserves the product before enclosure rather than maximizing its two factors independently, but it still pays every analytic remainder. It is an implementable certificate interface, not a claim that we have generated all the high-degree panel coefficients.

Proposition 5.3 (Why closeness of charts alone is insufficient)**.** *Let P, Q define charts by (2.1) on the same disk. Assume the P chart is a probability law on the domain considered, $\|P-Q\|_\infty \le d_0$, and $\|D^2(P-Q)\|_{\mathrm{op}} \le d_2$, including all rotations. Then, with the covariances associated to each chart,*

$$|\mathcal{L}_P p_i - \mathcal{L}_Q q_i| \le 18 d_2 + 60 d_0 \|D^2 q_i\|_{\mathrm{op}}. \tag{5.3}$$

For a common barrier ψ,

$$|(\mathcal{L}_P - \mathcal{L}_Q)\psi| \le 60 d_0 \|D^2\psi\|_{\mathrm{op}}.$$

Proof. Each grade coordinate lies in $[0,2]$; hence its variance $D_i \le 1$. Indeed $X^2 \le 2X$ implies $\mathrm{Var}(X) \le 2\mu - \mu^2 \le 1$. Positivity of C_P and (2.2) give $\operatorname{Tr} C_P = 6\sum_i D_i \le 18$, so its contraction with a Hessian of operator norm at most d_2 is at most $18 d_2$. The source's elementary covariance estimate is $\|\Delta C\|_{\mathrm{op}} \le 15 \max_i |\Delta D_i|$. For completeness, for $|e_i| \le 1$ its matrix has norm at most 15: the norm is convex in e, so test the eight sign vertices; equal signs give eigenvalues $9, 9$ up to sign, and a single opposite sign gives eigenvalues $15, -9$ up to sign. Here $\Delta D_i = 2(p_i - q_i)$, so $\|\Delta C\|_{\mathrm{op}} \le 30 d_0$; in dimension two its nuclear norm is at most $60 d_0$. Now split

$$\mathcal{L}_P p_i - \mathcal{L}_Q q_i = C_P : D^2(p_i - q_i) + (C_P - C_Q) : D^2 q_i.$$

The two norm bounds prove the assertions. □

With (2.6) and the source reference-Hessian bound 3.05, this gives only

$$18(4.327 \cdot 10^{-12}) + 183(3.517 \cdot 10^{-17}) < 7.790 \cdot 10^{-11}.$$

This upper bound is substantially above 10^{-13}. It does not show that the reference residual is actually large; it shows that these difference estimates alone cannot certify it at the improved tolerance. Replacing degree 780 by degree 420 would require a sharper direct residual proof and renewed geometry, boundary and interior checks. A tolerance gain is not a demonstrated speedup.

6 Rectangular witnesses retain their own premises

The rectangular part is a different construction, not a consequence of the square chart. A finite support Φ, a probability law p, and a readable label tree produce three charged edge rates (R_{XY}, R_{XZ}, R_{YZ}). The source's attainability theorem gives

$$W(R_{XY}, R_{XZ}, R_{YZ}) \le b \log 2,$$

for a base support with cost 2^b, where W is the homogeneous, coordinatewise nondecreasing rectangular exponent function. Label availability, completion, and zero cross terms are premises of this theorem; a numerical entropy triple alone does not prove it.

At a node incident to X, splitting into a child label L produces a charged term $\Pr(\nu) I(X; L \mid \nu)$. At a node not incident to X, the same entropy loss is uncharged. The chain rule, terminal determination of X, and telescoping give the exact source identity

$$H(X) - (R_{XY} + R_{XZ}) = \sum_{\nu \text{ not incident to } X} \Pr(\nu) I(X; L \mid \nu) \ge 0. \tag{6.1}$$

Thus a uniform X marginal and exact independence at every nonincident split force equality. Approximate equality is different.

Proposition 6.1 (Near-flatness pays exponent slack)**.** *Assume the stated tree attainability and its balanced convexification give $W(S/2, T, S/2) \le C$, where $S = R_{XY} + R_{XZ}$ and $C = b \log 2 > 0$.*

Suppose $a, \ell \geq 0$, $H(X) \geq C - a$, the right side of (6.1) is at most ℓ, and $T \geq T_0 \geq 0$, with $D = a + \ell < C$. Then, for

$$k_0 = \min\{1,\, 2T_0/(C - D)\}, \qquad w(k_0) \leq \frac{2C}{C - D}. \tag{6.2}$$

Positive D does not by itself certify $w(k_0) = 2$.

Proof. Equation (6.1) gives $S \geq C - D > 0$. Monotonicity reduces the two equal edges to $(C - D)/2$ and the middle edge to T_0. Homogeneity rescales these to $(1, 2T_0/(C - D), 1)$, with cost at most $2C/(C - D)$. If the middle ratio exceeds one, a further monotone restriction gives $k_0 = 1$. The final assertion follows because the displayed bound is strictly larger than two for $D > 0$. □

For instance $C = 1, D = 1/100, T_0 = 23/100$ gives $k_0 = 46/99$ and the certified upper bound $200/99$, not two. These are manufactured rate budgets, not a new tensor witness. They quantify why numerical near-flatness cannot replace the source's symbolic equality for its dual bound.

The source's special finite fiber construction has sizes $1, 2, 4, 19$ with multiplicities $8, 12, 6, 1$. Its tilted law and entropy formulas use these exact multiplicities; the existence of the $(19, 4, 1)$ support, the readable extraction steps and their charged cost remain separate. We fetched and literally inspected the pinned original `verify_rational.py` (141 lines) and `verify_construction.py` (83 lines), then ran both in an owned directory. Each exited zero, and its parsed JSON matched the pinned reference output. The former verifies rational logarithm and finite-value inequalities; the latter finite ownership, completion, fiber and five-step counts. Neither checks the full tensor attainability theorem or the whole rectangular baseline table.

7 Verification, limits, and research directions

7.1 What was and was not verified

All 17 cached scientific TeX files, the bibliography and the available source/verification text documentation were literally read in bounded displays. This includes the algebra, attainable values, AP compilation, W seed, complete square certificate and all rectangular scientific modules, not only an introduction and rank lemma. All 299 lines of one original private clearance chapter were freshly read. Referenced papers, the remainder of the private corpus, the separate October 2 companion proof and Lean developments are not covered by that statement. Before/after hashes and actual intervals are recorded in the owned reading ledger.

The source full square package requires a POSIX process environment, GMP, C++ integer support and its prescribed undefined-behaviour sanitizer check. That complete computation was not executed here. The byte-count/SHA inventory script is not a numerical proof and is not reported as one. The imported square bounds in (2.6) and (4.1), the sample-to-domain enclosures, and the source's finite tensor realizability assertions are therefore explicit premises of our quantitative instantiation. No arithmetic or logical defect is alleged merely because their complete independent re-execution is outside this report's scope.

The independent author script `M217-checks.py` uses exact integer/Fraction arithmetic for covariance and rounding identities, Bernstein restriction identities, product-error bounds, the full three-budget computation, the frontier and rectangular slack. Its receipt reports actual case and assertion counts by category. The final run passed 1,085 cases and 4,255 assertions: 462 probability laws, eight covariance sign vertices, 154 small-degree operator coefficients, 244 Bernstein panel evaluations, four positivity panels, 72 product-error cases, one combined source-budget certificate, 20 frontier cases and 120 rectangular rate budgets. These finite checks supplement the proofs, do not certify untested degrees or every panel, and exclude both parent checks and the two original-checker replays.

7.2 Advantages and costs

The signed test charges only the harmful residual direction. Its denominator-free form can exploit cancellation on a whole panel. The inset theorem specifies the usable η-window, instead of silently optimizing an asymptotic limit. The two correction parameters make boundary utility, interior curvature and central excess separately auditable. Their exact frontier says how much a simplified future chart would have to achieve before it is worth regenerating expensive certificates.

These advantages have real costs. The correction deliberately reduces central utility, so it does not improve the exponent. Large-degree product polynomials may cost more to enclose than separate low-degree factors. Bernstein positivity is sufficient, not necessary at a fixed subdivision depth. Chart perturbations change the covariance, domain, active faces and boundary laws; no one residual norm controls all four. The six-state supergradient relaxation may be much larger than the true set of supporting covectors. A sharpened covector polytope would need additional proved attainable-value information.

7.3 Feasible extensions and contribution assessment

A useful next experiment would generate a moderate-degree exact chart, certify its residual directly, and then recheck every changed geometric and boundary input against (4.4). Until that work is done, there is no runtime or degree reduction claim. Another route is a nonradial polynomial correction χ with a separately certified boundary sign, centre value and $\mathcal{L}\chi$; our maximum argument applies unchanged, but solving the finite design problem is not the certificate. Panelwise use of (3.5) can combine these approaches. For rectangular witnesses, retaining exact symbolic independence while optimizing the remaining entropy directions is preferable to treating a small numerical entropy deficit as zero.

The applicability remains L3. The contribution is assessed C3 because the source-calibrated uniform inset and three-budget frontier constitute a concrete local strengthening of its comparison certificate, with a proved and exactly checked tolerance gain and a countervailing chart-transfer cost. The linear-programming, Bernstein, covariance, entropy and maximum principles are classical and are not credited as new theories. No new best exponent, all-field algorithm, globally verified tensor proof, independently established novelty priority or practical complexity improvement is claimed.

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